

PRESIDENTIAL GRADUATE SCHOOL

Course: Knowledge Discovery and Data Science (BSIT-375)

Week 8: Final Project Assignment

Nepal-Pulse

*A Distributed Real-Time Knowledge Discovery Engine
for Multi-Lingual News Streams*

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Project Overview

This project challenges you to design and implement a **real-time news intelligence system** focused on answering the question: “**What is happening in Nepal?**” You will build a streaming data pipeline that scrapes at least 20 top Nepali and English news portals, processes the content using PySpark, extracts emerging topics via Machine Learning (LDA, NMF, or BERTopic), and presents insights through an interactive dashboard. The system must be scalable, containerised, and capable of updating in near real-time.

1 Project Objectives

By completing this assignment, students will:

- Design and implement a **polling-based ingestion service** in TypeScript to scrape the home-pages of **20 major news portals** (10 Nepali + 10 English) every 5 minutes.
- Build a **streaming data pipeline** using Apache Kafka to decouple ingestion, processing, and model inference.
- Apply **PySpark Structured Streaming** for multi-lingual text preprocessing, cleaning, and windowed transformations.
- Develop a **Real-Time ML Service** implementing at least one topic modeling algorithm:
 - **LDA** (Latent Dirichlet Allocation)
 - **NMF** (Non-Negative Matrix Factorization)
 - **BERTopic** (Transformer-based Topic Modeling)
- Create an **interactive dashboard** (Grafana, Streamlit, or Custom UI) to visualize trending topics, entity frequencies, and topic evolution over time.
- Containerise all services with Docker and orchestrate using Docker Compose, with optional distributed deployment across multiple physical nodes.

2 System Architecture Requirements

The pipeline follows a layered, event-driven design (Figure 1). Communication between services is asynchronous via Kafka topics.

2.1 Data Ingestion Service (Polling Layer)

- **Language:** TypeScript (Node.js).
- **Responsibilities:**
 - a. **Scraping Engine:** Periodically fetch HTML from 20 portals. Extract headlines, summaries, and categories.
 - b. **Bilingual Support:** Ensure proper handling of both English and Nepali (Unicode) characters.
 - c. Publish raw data as JSON to Kafka topic `raw-news`.
 - d. Implement robust error handling and rate-limiting to prevent IP blocking.

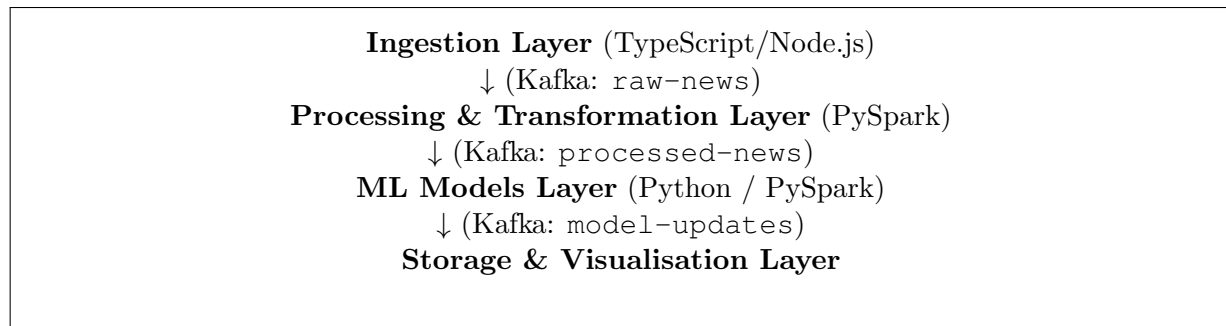


Figure 1: High-level pipeline architecture

2.2 Data Preprocessing & Transformation Layer

- **Language & Framework:** Python with **PySpark** (Structured Streaming).
- **Responsibilities:**
 - a. Consume messages from `raw-news`.
 - b. **Text Cleaning:** Remove HTML tags, normalize Unicode, and filter stop-words for both languages.
 - c. **NER (Named Entity Recognition):** Extract key Persons, Organizations, and Locations from headlines.
 - d. Publish cleaned and enriched data to Kafka topic `processed-news`.

2.3 ML Models Service (Topic Discovery Layer)

- **Language & Framework:** Python (PySpark MLlib, Gensim, or BERTopic).
- **Responsibilities:**
 - a. Implement **LDA, NMF, or BERTopic** to discover latent themes in the news stream.
 - b. Generate topic distributions for each new article batch.
 - c. Publish topic keywords and document-topic assignments to `model-updates`.

2.4 Storage & Visualisation Layer

- **Storage:** PostgreSQL or a time-series optimized DB (e.g., InfluxDB).
- **Dashboard Requirements:**
 - Top 10 Trending Topics (Word Clouds or Bar Charts).
 - Topic Evolution over time (Line/Area charts).
 - Entity Frequency (Most mentioned People/Organizations in Nepal).
 - Source Distribution (Comparison of news volume across portals).
 - The dashboard must refresh automatically to reflect live news updates.

3 Technology Stack & Constraints

Component	Allowed Technologies	Notes
Ingestion	Node.js / TypeScript	Must be containerised
Message Broker	Apache Kafka	Use Confluent Docker images
Processing	PySpark (Python)	Structured Streaming
ML Models	Python / PySpark	LDA, NMF, or BERTopic
Database	PostgreSQL / InfluxDB / Elasticsearch	Time-series or Document store
Visualisation	Grafana / Streamlit / Custom React UI	Real-time updates required
Orchestration	Docker Compose	Single <code>docker-compose.yml</code>

Table 1: Technology stack and constraints

4 Project Presentation Guidelines

1. **Source Code:** A GitHub repository containing the complete codebase.
2. **Documentation:** A README file describing setup, configuration, and architecture.
3. **Demonstration:** A live demo showing data flowing from news portals to the dashboard.

May your pipeline flow without backpressure and your topics be insightful!